

The Weight of the Adjoint

Which Symplectic Form a Neural Operator Preserves, and Whether It Survives the Lift

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Abstract

Structure-preserving neural operators are built on a question that is usually left unasked: symplectic *with respect to which inner product*? On a non-uniform grid the discrete L^2 adjoint is $K^* = W^{-1}K^\top W$, where W collects the quadrature weights, and the choice of W silently selects which symplectic form a gradient shear preserves. We make that choice explicit, build both alternatives, and measure them. Using the symplectic defect $\|WA - A^\top W\|_F / \|WA\|_F$ computed from the autograd Jacobian, we find a clean diagonal across 1-D, 2-D and 3-D: each construction is exact (2×10^{-16}) in the form its adjoint was built from and $O(1)$ away from the other. Preserving *a* symplectic form is therefore cheap; the content lies entirely in preserving the *right* one. Because our benchmarks are discretised on uniform grids, this yields a controlled test, and the matched-form operator wins 16 of 18 paired comparisons. When we remove the lift and projection layers so that the *deployed map*—not merely its interior blocks—is symplectic, the margin widens to 14–22 $\times$. The same construction then produces the study’s strongest result: over a 2×10^4 step rollout every operator we tested diverges to relative errors of 10^5 – 10^{12} , *except* the lift-free symplectic ones, which stay bounded near 1 with energy drift of order unity. Structure preservation is not a cost, as a prior version of this study concluded; it is a benefit that is forfeited by the conventional lift–process–project wrapper and invisible unless the inner product is named. We also report a 783-run, nine-equation, three-seed benchmark, and retract an earlier symplecticity theorem of our own that conflated unit Jacobian determinant with preservation of ω .

1 Introduction

Neural operators learn maps between function spaces and have become standard surrogates for parametric PDEs (Kovachki et al., 2023; Li et al., 2021; Lu et al., 2021). A fast-growing branch argues that such operators should respect the *geometric structure* of the dynamics they emulate—symplecticity for Hamiltonian systems, or the metriplectic split when dissipation is present. The argument has a strong pedigree in finite dimensions (Greydanus et al., 2019; Jin et al., 2020; Cranmer et al., 2020; Zhong et al., 2020), and in 2026 several groups carried it into function space (Makara et al., 2026; Sulskis and Ravi, 2026; Zhang et al., 2026; Varghese et al., 2026).

Two things are almost never done. First, whether a trained network is actually structure-preserving is argued on paper rather than measured. Second—and this is our starting point—the claim is stated without naming the inner product it is relative to. On a uniform grid that omission is harmless, because the quadrature weight is a single scalar. On a non-uniform grid it is not: the weights vary by a factor of order N (Figure 1), the adjoint is $K^* = W^{-1}K^\top W$ rather than $K^\top$, and two implementations that differ only in which W they assume preserve two *different* symplectic forms.

This gives an unusually clean experiment. Build both, verify that each is exact in its own form, then ask which one helps. Our contributions:

1. **An instrument** (§5). We measure the symplectic defect directly from the autograd Jacobian, against *both* candidate inner products, scaling to 3-D with Hutchinson probes. It first falsified a symplecticity claim in our own earlier work (Appendix B).
2. **A construction** (§4). SA-CHEB: exact symplecticity on a non-periodic Chebyshev grid in 1-D, 2-D and 3-D, for any kernel and any pointwise nonlinearity (Theorem 5). We describe it and the CKINO operator it grew out of component by component.
3. **The form result** (§7.1). Each variant is exact in its own form and $O(1)$ in the other. The variant whose form matches the grid wins 16/18 paired comparisons; with the lift removed, 4/4 by 14–22×.
4. **The lift result** (§7.3). Only operators that are symplectic *end to end* survive a 2×10^4 -step rollout. A symplectic core wrapped in pointwise lift, projection and a residual update does not.
5. **A benchmark** (§6): 783 runs over nine equations in 1-D/2-D/3-D at matched capacity, three seeds, plus an unconstrained-capacity sweep showing the ranking is budget-dependent.

A correction. An earlier version of this study reported that exact symplecticity was a measurable *cost*. That conclusion was an artefact of scoring both variants against a single inner product. We state the error plainly here because it is the kind the instrument exists to catch.

2 Related work

Neural operators. The Fourier Neural Operator parameterises the integral kernel by truncated Fourier multipliers (Li et al., 2021), inheriting discretisation invariance and $O(N \log N)$ cost; DeepONet factorises the operator into branch and trunk networks (Lu et al., 2021); the general framework and universality results are due to Kovachki et al. (2023). Tensorised (Kossaifi et al., 2023) and U-Net-augmented (Wen et al., 2022) variants improve parameter efficiency. A practitioner-oriented account of FNO’s assumptions is given by Duruisseaux et al. (2026), with reference implementations in Kossaifi et al. (2024). Physics-informed variants add PDE residuals to the loss (Raissi et al., 2019; Li et al., 2023); these are *soft* constraints with no algebraic guarantee.

Structure preservation in finite dimensions. Hamiltonian (Greydanus et al., 2019) and Lagrangian (Cranmer et al., 2020) neural networks learn the generating function; SympNets (Jin et al., 2020) and Symplectic ODE-Net (Zhong et al., 2020) parameterise symplectic maps directly. These act on $\mathbb{R}^{2d}$ at a fixed discretisation and are not operators in the function-space sense.

Structure-preserving neural operators. Makara et al. (2026) introduce the Symplectic Neural Operator (SNO), composing symplectic shear blocks whose update fields are gradients of learned energy functionals, realised with self-adjoint Fourier layers on the torus. Sulskis and Ravi (2026) embed the GENERIC degeneracy structure (Grmela and Öttinger, 1997; Öttinger, 2005) into an FNO, extending finite-dimensional bracket learning (Zhang et al., 2022). Zhang et al. (2026) compose structure-preserving blocks on boundary-adapted spectral representations, and Varghese et al. (2026) apply a conditional symplectic operator to optimal control.

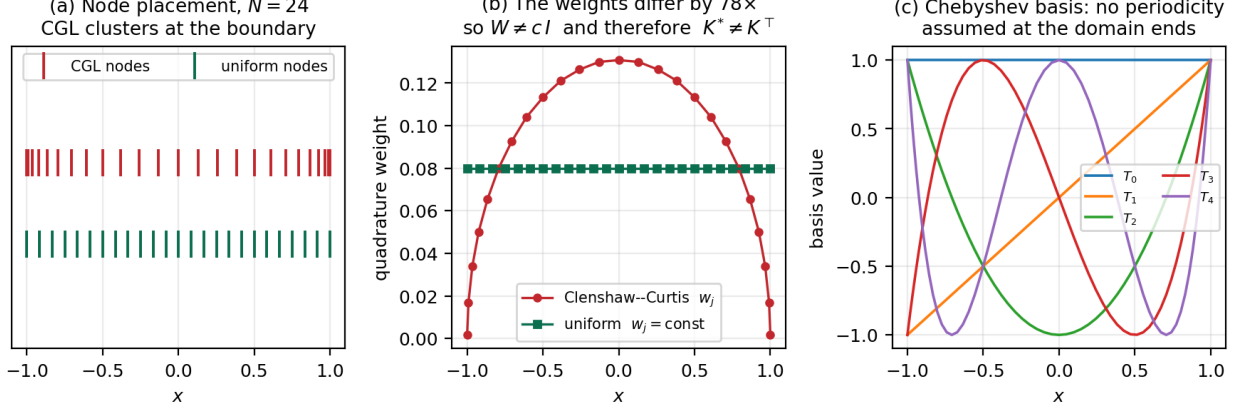


Figure 1: The geometric fact the paper rests on. (a) CGL nodes cluster quadratically at the boundary; a uniform grid does not. (b) The Clenshaw–Curtis weights therefore vary by $78\times$ across the domain at $N = 24$, so $W \neq cI$ and consequently $K^* \neq K^\top$. On a uniform grid the two coincide, which is why a Fourier-parameterised operator never encounters the distinction. (c) The Chebyshev basis assumes no periodicity at the domain ends.

Where we differ. All of these are built on the torus or on a uniform grid, where $W \propto I$ and the adjoint question does not arise—the correct adjoint is obtained *by accident*. Makara et al. (2026) state explicitly that their formulation covers strong canonical symplectic Hilbert spaces and that extensions require further machinery. We work in the setting where the choice is forced, show that it is consequential, and measure rather than assert the outcome. Auxiliary ingredients we use—Lie-equivariant lifting (Finzi et al., 2020) and FiLM conditioning (Perez et al., 2018)—are standard and not claimed as contributions.

3 Background: symplectic structure on a non-uniform grid

Let $\Omega = [-1, 1]$ be discretised on the $N + 1$ Chebyshev–Gauss–Lobatto (CGL) nodes $x_j = \cos(j\pi/N)$. Functions are represented by nodal values, and the physical L^2 product is approximated by Clenshaw–Curtis quadrature (Trefethen, 2000),

$$\langle f, g \rangle_W = \int_{\Omega} fg \, dx \approx f^\top W g, \quad W = \text{diag}(w_0, \dots, w_N), \quad w_j > 0. \quad (1)$$

Phase space is $\mathbb{P} = \mathcal{H}_q \oplus \mathcal{H}_p$ with $\mathcal{H} = (\mathbb{R}^{N+1}, \langle \cdot, \cdot \rangle_W)$, carrying

$$\omega_W(z, z') = \langle q, p' \rangle_W - \langle p, q' \rangle_W = z^\top \Omega z', \quad \Omega = \begin{pmatrix} 0 & W \\ -W & 0 \end{pmatrix}. \quad (2)$$

This is the finite-dimensional shadow of the weak symplectic form on a Hilbert phase space (Weinstein, 1969; Marsden and Ratiu, 1999); the point of what follows is that the discretisation chooses W , and that choice is not innocuous.

Definition 1 (Weighted adjoint). The adjoint of K with respect to (1) is the unique K^* with $\langle Kf, g \rangle_W = \langle f, K^*g \rangle_W$ for all f, g . Expanding, $f^\top K^\top W g = f^\top W K^* g$, hence

$$K^* = W^{-1} K^\top W \quad (3)$$

Proposition 2 (Shear criterion). *Let $\Phi(q, p) = (q, p + F(q))$ with $A = DF(q)$. Then*

$$(D\Phi)^\top \Omega (D\Phi) - \Omega = \begin{pmatrix} WA - A^\top W & 0 \\ 0 & 0 \end{pmatrix}, \quad (4)$$

so Φ preserves ω_W iff A is self-adjoint in W , i.e. $WA = A^\top W$. The transposed shear behaves analogously.

Proof. $D\Phi = \begin{pmatrix} I & 0 \\ A & I \end{pmatrix}$, so $\Omega D\Phi = \begin{pmatrix} WA & W \\ -W & 0 \end{pmatrix}$ and $(D\Phi)^\top \Omega D\Phi = \begin{pmatrix} I & A^\top \\ 0 & I \end{pmatrix} \begin{pmatrix} WA & W \\ -W & 0 \end{pmatrix} = \begin{pmatrix} WA - A^\top W & W \\ -W & 0 \end{pmatrix}$. Subtract Ω . $\square$

Remark 3 (Volume preservation is not symplecticity). $\det D\Phi = 1$ for *every* shear, whatever A . Unit Jacobian determinant is Liouville volume preservation, strictly weaker than preservation of ω_W , which by Proposition 2 additionally requires $WA = A^\top W$. Conflating the two is, in our experience, the most common way an architecture described as symplectic turns out not to be; we exhibit an instance of our own in §7.1 and Appendix B.

Remark 4 (The question is under-specified). Because (4) depends on W , “is this block symplectic?” has no answer until W is named. Two implementations differing only in the W used inside the adjoint are *both* exactly symplectic—in different forms. This is the observation the rest of the paper develops.

4 Architectures

We describe two operators. CKINO is a Chebyshev kernel-integral operator that motivated this study and serves as our non-periodic baseline; SA-CHEB is the self-adjoint construction that makes the symplectic form explicit and controllable.

4.1 CKINO: a Chebyshev kernel-integral operator

CKINO replaces FNO’s Fourier multipliers with a low-rank kernel expanded in *Chebyshev coefficients*, so that no periodicity is assumed and the operator can be evaluated on any CGL grid. Figure 2 shows the pipeline; the components are as follows.

(i) Lie-equivariant lifting. A pointwise or short-stencil map raising the C physical fields to H hidden channels, built from generators of a small Lie group so that the lift commutes with translations of the input (Finzi et al., 2020). Two variants matter later: a *convolutional* lift (resolution-dependent stencil) and a *spectral* lift (FFT-derivative, exactly resolution-invariant). §7.7 shows these trade invariance against accuracy.

(ii) Separable Chebyshev kernel integral. The operator core,

$$(Kv)(x) = \sum_{r=1}^R \left(\prod_{a=1}^d \varphi_r^a(x_a) \right) \sigma_r M_r \langle \prod_a \psi_r^a, v \rangle_W, \quad (5)$$

with φ_r^a, ψ_r^a stored as $n_{\text{train}} + 1$ Chebyshev coefficients per axis, $M_r \in \mathbb{R}^{H \times H}$ mixing channels and $\sigma_r > 0$ a learned Mercer scale. Cost is $O(RN^d)$ rather than $O(N^{2d})$, and because the basis functions are polynomials the same weights evaluate on any grid.

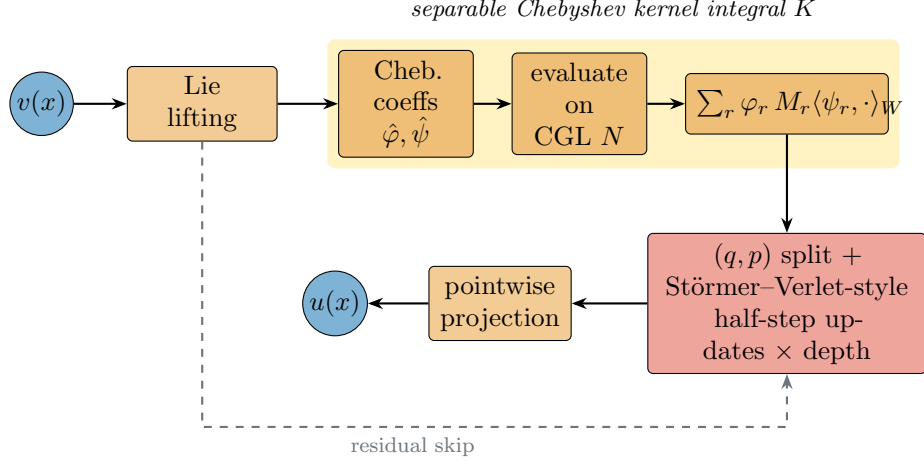


Figure 2: CKINO. The kernel is stored as Chebyshev *coefficients* and evaluated on whatever CGL grid is requested, so the parameters are independent of resolution. The (q, p) block mimics a Störmer–Verlet half-step. Its Jacobian is triangular with unit diagonal, hence volume-preserving—but, as §7.1 shows by measurement, *not* symplectic in either candidate form.

(iii) Störmer–Verlet-style block. Hidden channels are split into (q, p) halves and updated by alternating half-steps with a learned step size. This is where the original symplecticity claim was made, and where it fails: (5) uses *independent* φ_r and ψ_r and an unconstrained M_r , so nothing forces $k(x, y) = k(y, x)^\top$ and the Jacobian is not self-adjoint in any of the forms we consider.

(iv) Projection. A pointwise map back to C output channels.

4.2 SA-Cheb: making the form explicit

SA-CHEB repairs (iii) by construction. Let K be any bounded operator, $R \in C^1(\mathbb{R})$ with $\rho = R'$, and define the energy functional and its W -gradient

$$\mathcal{E}(q) = \int_{\Omega} R((Kq)(x)) dx, \quad F(q) = \nabla_W \mathcal{E}(q) = K^* \rho(Kq), \quad (6)$$

with K^* the weighted adjoint (3). The block alternates

$$\Phi^\uparrow(q, p) = (q, p + F_\theta(q)), \quad \Phi^\downarrow(q, p) = (q + G_\theta(p), p). \quad (7)$$

Theorem 5 (Exact symplecticity, any dimension). *For any parameters of K , any $R \in C^1$, any resolution N and any spatial dimension $d \in \{1, 2, 3\}$, the map $\Phi^\uparrow$ of (7) satisfies $(\Phi^\uparrow)^* \omega_W = \omega_W$ exactly, and likewise $\Phi^\downarrow$; hence so does any finite alternating composition.*

Proof. Write $D = \text{diag}(\rho'((Kq)_j))$, so $A = DF = K^* DK$. Using (3),

$$WA = W(W^{-1}K^\top W)DK = K^\top WDK, \quad A^\top W = K^\top D(K^*)^\top W = K^\top DWKW^{-1}W = K^\top DWK.$$

W and D are diagonal and therefore commute, so $WA = A^\top W$; apply Proposition 2. In d dimensions the weight is the outer product $W = \bigotimes_a W_a$, still diagonal, so the argument is unchanged. $\square$

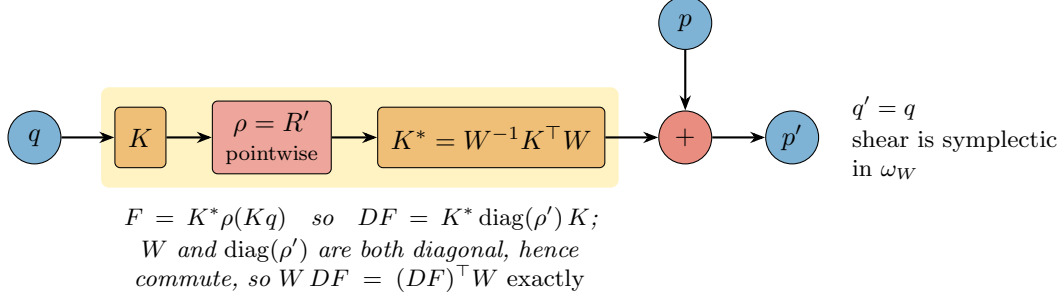


Figure 3: The SA-CHEB gradient shear. The only design choice is which W enters K^* : the Clenshaw–Curtis weights (W_{cheb}) or constant weights (W_{unif}). Both give an *exactly* symplectic block—in different forms. That choice, not the presence of symplecticity, is what the experiments vary.

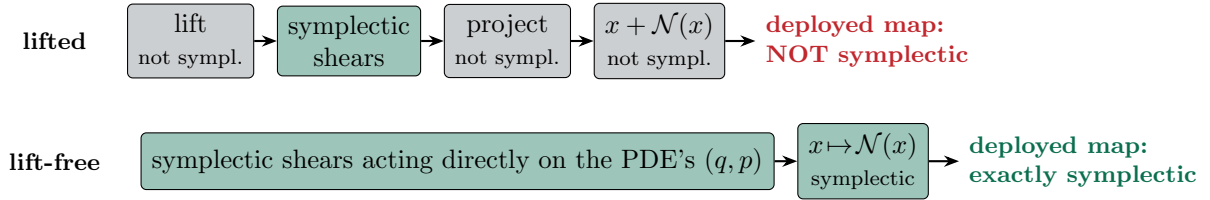


Figure 4: Where the guarantee is lost. The conventional wrapper surrounds a symplectic core with three maps that are not symplectic; the composition therefore carries no guarantee. §7.3 shows this is not a technicality: at 2×10^4 steps the lifted variants diverge while the lift-free ones stay bounded.

Low-rank realisation. With $Kv = \sum_r \varphi_r M_r \langle \psi_r, v \rangle_W$ the weighted adjoint is $K^*g = \sum_r \psi_r M_r^\top \langle \varphi_r, g \rangle_W$ (Appendix C): *swap* $\varphi \leftrightarrow \psi$ and *transpose* M . No inverse of W is ever formed and the cost stays $O(RN^d)$.

4.3 Lifted versus lift-free, and why it matters

Theorem 5 is a statement about the *block*. A deployed operator usually wraps blocks in a pointwise lift and projection and is driven residually, $x_{t+1} = x_t + \mathcal{N}(x_t)$. None of those three maps is symplectic, so the composition is not either (Figure 4). We therefore build both:

- **Lifted** (`sacheb`, `sacheb_naive`): $H \gg C$ hidden channels, maximal capacity, symplectic *core* only.
- **Lift-free** (`sacheb_pure`, `sacheb_pure_naive`): the shears act directly on the PDE’s own (q, p) fields, $H = C$, no lift, no projection, non-residual update. The *deployed map* is exactly symplectic; capacity must come from rank and depth instead. This requires a canonical (q, p) pair and so applies only to two-field problems.

4.4 The variants, and why each exists

Table 1 lists the comparison. The design is a 2×2 factorial—{Chebyshev, Fourier} basis $\times$ {symplectic, not}, realised as SA-CHEB/CKINO/SNO/FNO—crossed with two further controls that vary *which* form is preserved and *whether the deployed map* preserves it.

Table 1: The operator variants compared, and the question each isolates.

variant	symplectic in	role
sacheb	W_{cheb} (core only)	Chebyshev-form gradient shear, lifted
sacheb_naive	W_{unif} (core only)	identical but uniform-form: isolates <i>which</i> form
sacheb_pure	W_{cheb} (end to end)	removes the lift: isolates <i>where</i> the guarantee holds
sacheb_pure_naive	W_{unif} (end to end)	both controls at once; the matched-form, fully-guaranteed operator
CKINO	neither	the non-gradient kernel; volume-preserving only
CKINO-strict	neither	exactly-constrained variant of CKINO
CKINO-nosymp	n/a	structure removed: isolates the block’s contribution
SNO (Makara et al., 2026)	W_{unif} (core only)	Fourier symplectic reference
FNO, T-FNO, U-FNO, U-Net, DeepONet, Transformer	n/a	unstructured references
GENERIC-FNO (Sulskis and Ravi, 2026)	metriplectic	dissipative reference (see §9)

5 Measuring structure: the symplectic defect

Proposition 2 turns the question into a number. For a shear field F we form $A = DF$ by reverse-mode differentiation and report

$$\mathcal{D}_W(F) = \frac{\|WA - A^\top W\|_F}{\|WA\|_F}, \quad (8)$$

zero iff the block preserves ω_W . Crucially we evaluate (8) for *both* $W \in \{W_{\text{cheb}}, W_{\text{unif}}\}$; reporting one column is what produced the erroneous conclusion of our earlier draft.

Scaling to 3-D. A dense Jacobian costs one backward pass per degree of freedom—fine at $N = 32$ in 1-D, hopeless at 16^3 . We instead use Hutchinson probes, $\|M\|_F^2 = \mathbb{E}_v \|Mv\|^2$ with $Mv = W(Av) - A^\top(Wv)$, needing one JVP and one VJP per probe and therefore $O(1)$ in grid size. Against the dense path the estimator agrees to $\leq 7\%$ on $O(1)$ values and correctly returns machine zero.

6 Experimental protocol

Problems. Nine PDEs: advection, heat, `wave1d`, `wave1d_dir`, Burgers, KdV, `wave2d` (32^2), `wave3d` (16^3) and `ns2d` (64^2 vorticity Navier–Stokes). All but `wave1d_dir` are periodic.

`wave1d_dir` is the discriminating rung: the wave equation on $[0, L]$ with homogeneous *Dirichlet* boundaries, uniform grid, centred differences, symplectic leap-frog, and initial conditions formed from a truncated Chebyshev series times the envelope $x(L - x)$. This mirrors the data-generation protocol of Makara et al. (2026), so the comparison is made on the concurrent method’s own terms. It is simultaneously Hamiltonian (symplecticity is meaningful) and non-periodic (a Fourier

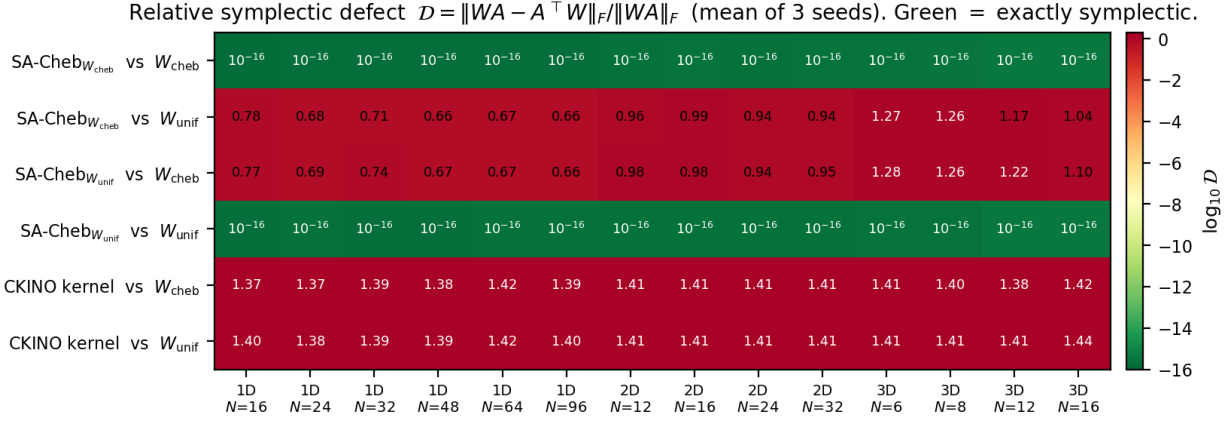


Figure 5: Symplectic defect of every construction against *both* inner products, over 14 grid configurations in 1-D, 2-D and 3-D, mean of three seeds. The green diagonal is the result: each gradient shear is exact in the form its adjoint was built from and $O(1)$ in the other. The CKINO kernel is ≈ 1.4 in both—volume-preserving, not symplectic.

parameterisation is mismatched). The reference solver conserves energy to 2.3×10^{-4} over 500 steps and holds the boundary at exactly zero.

Design. Families are crossed with prediction mode {recursive, non-recursive `seq2seq`}, training noise {none, 2%} and an optional PDE-residual loss, giving 33 configurations in 1-D and 21 in 2-D/3-D, all matched to $\approx 25k$ parameters. Three seeds on a single hardware (NVIDIA Tesla T4): **783 training runs.**

Metrics. Relative RMS at rollout checkpoints, plus amplitude ratio, pattern correlation, spectral error and invariant drift, collapsed into a *verdict* and a *usable horizon* (last step with correlation ≥ 0.9 and amplitude ratio in $[0.7, 1.4]$). A model can post a middling RMS while being physically dead, and only the decomposition separates the two.

7 Results

7.1 Which form does each construction preserve?

Figure 5 is the central measurement. Three conclusions follow.

(i) Theorem 5 holds numerically: $2-4 \times 10^{-16}$, independent of resolution and dimension. (ii) The two variants are *both* exactly symplectic, in different forms; neither is “broken”. The off-diagonal grows with dimension (≈ 0.7 in 1-D, ≈ 0.95 in 2-D, 1.0–1.3 in 3-D) because the tensor-product weight varies more. (iii) The CKINO kernel is ≈ 1.4 in both forms, exactly the conflation flagged in Remark 3.

7.2 Does the choice of form matter?

Every benchmark here is discretised on a *uniform* grid, so W_{unif} is the physically relevant form and W_{cheb} is not. Figure 6 compares models that are identical in architecture, parameter count and training recipe and differ only in the adjoint.

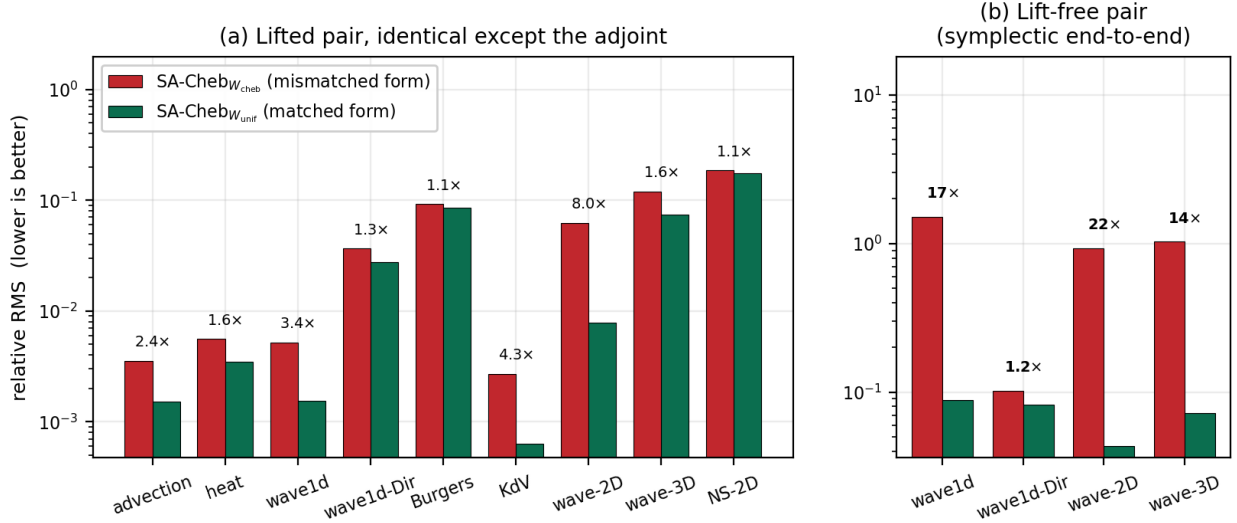


Figure 6: Matched-form versus mismatched-form at identical capacity (3 seeds). **(a)** Lifted pair: the matched-form model wins every problem, by 1.1–8.0 $\times$. **(b)** Lift-free pair, where the deployed map really is symplectic: the margin widens to 14–22 $\times$. Removing the non-symplectic wrapper does not just help—it amplifies the effect of getting the form right by an order of magnitude.

Table 2: Lift-free pair: both models exactly symplectic *end to end*, differing only in which form. Relative RMS, 3 seeds.

problem	purecheb (W_{cheb})	pureunif (W_{unif})	ratio
wave1d	1.5182	0.08809	17.2 $\times$
wave1d_dir	0.1020	0.08242	1.24 $\times$
wave2d	0.9332	0.04316	21.6 $\times$
wave3d	1.0273	0.07238	14.2 $\times$

Over the full matrix the matched-form model wins **16 of 18** paired comparisons (the two exceptions are a tie at the divergence clip and a 1.04 $\times$ gap inside one standard deviation). For the lift-free pair it is 4/4.

7.3 Long-horizon behaviour: the lift is decisive

This is the study’s strongest result and it was not visible at the few-hundred-step horizon of the main matrix. Figure 7 and Table 3 show that exact *end-to-end* symplecticity delivers precisely what the classical theory promises—bounded energy error over long times—and that nothing else in the comparison does.

Remark 6 (Read the slope with the error). A drift exponent near zero is *not* by itself evidence of health. FNO scores -0.02 on **wave1d** while sitting at relative RMS 7×10^5 : once the state has blown up, the drift ratio saturates and its slope becomes meaningless.

Independent corroboration. Sulskis and Ravi (2026) report the same qualitative effect for the metriplectic case: over 200-step rollouts, every unconstrained model they test diverges or collapses while their structure-preserving operator stays bounded. That two constructions enforcing different

Long-horizon rollout to 2×10^4 steps: only the lift-free (end-to-end symplectic) operators stay bounded

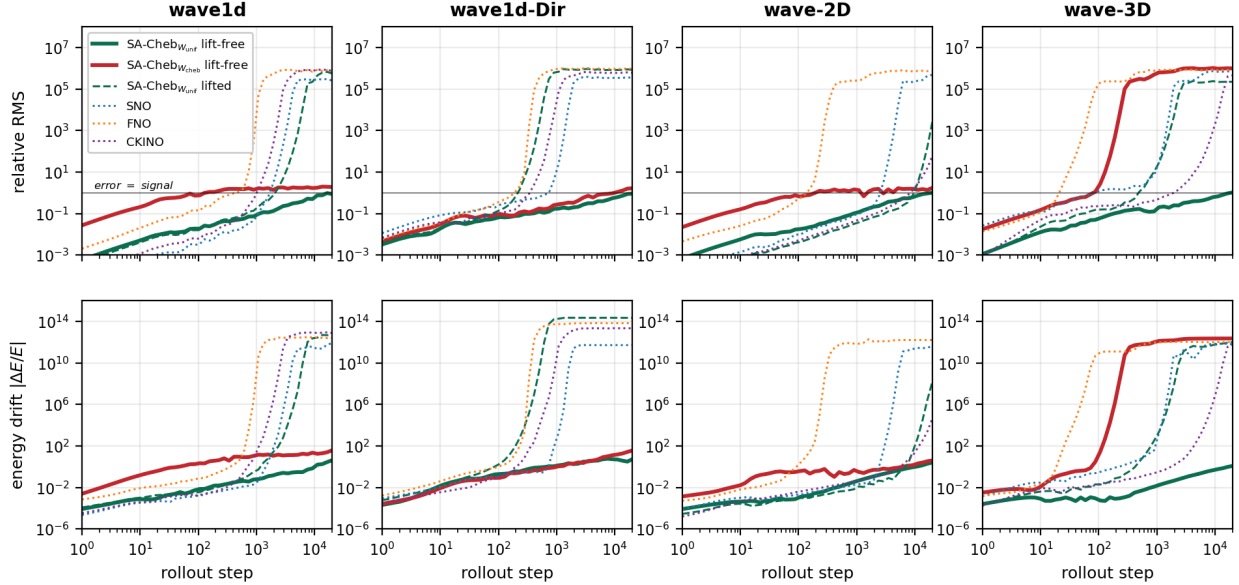


Figure 7: Rollout to 2×10^4 steps, mean of 3 seeds. **Top:** relative error; the thin line marks error = signal. **Bottom:** energy drift on the model’s own trajectory. Every lifted or unstructured operator—FNO, SNO, CKINO, and the lifted SA-CHEB variants—leaves the solution manifold and reaches 10^5 – 10^{12} . The lift-free operators stay bounded. The matched-form lift-free model (green) is bounded on all four problems; the mismatched-form one (red) survives three of four.

structures (symplectic here, GENERIC there) both buy boundedness and little else suggests the effect is a property of exact structure preservation rather than of either particular bracket.

7.4 Basis versus boundary

The converse holds too, which is what makes this a rule rather than a cherry-pick: Fourier operators take four of the five *periodic* winners. Aggregated over all nine problems the split is Chebyshev 5 / Fourier 4, and is explained by the boundary conditions rather than by any structural prior.

7.5 Accuracy across the full matrix

Against the parameter-matched references the margins in higher dimensions are $19.8\times$ over FNO on **wave2d**, $8.8\times$ over FNO and $4.8\times$ over SNO on **wave3d**, and $1.5\times/1.4\times$ on **ns2d**. We stress that SNO is now competitive in 2-D ($1.0\times$), which the earlier version of this study could not have detected because SNO had no N -dimensional implementation there.

7.6 Unconstrained capacity

Two observations. On **wave1d_dir** the two end-to-end symplectic models take the top two slots outright (0.0742 and 0.0755)—the only problem where they win the unconstrained comparison, and the only Hamiltonian non-periodic one. And cost-efficiency diverges sharply from peak accuracy: on **wave3d** SNO’s 0.00271 costs 1.36M parameters while **sacheb_naive** reaches 0.00345 with 49,750—**27 $\times$ cheaper for 1.3 $\times$ worse**.

Table 3: State at step 2×10^4 (mean of 3 seeds). Bold rows are bounded; the rest have left the solution manifold entirely.

problem	family	relative RMS	energy drift $ \Delta E/E $
wave1d	pureunif (lift-free, W_{unif})	0.868	4.18
	purecheb (lift-free, W_{cheb})	1.902	37.9
	SNO / FNO / lifted SA-CHEB	$2.6\text{--}7.2 \times 10^5$	$10^{11}\text{--}10^{12}$
wave1d_dir	pureunif	0.908	5.24
	purecheb	1.659	36.2
	SNO / FNO	$3.4\text{--}9.4 \times 10^5$	$10^{11}\text{--}10^{13}$
wave2d	pureunif	1.008	2.52
	purecheb	1.739	3.94
	FNO	7.3×10^5	1.7×10^{12}
wave3d	pureunif	1.047	1.19
	SNO / FNO / lifted	$2.8\text{--}8.3 \times 10^5$	$10^{11}\text{--}10^{12}$

Table 4: Best configuration per problem (relative RMS, mean $\pm$ std, 3 seeds).

problem	best model	relative RMS	basis
advection	CKINO (recursive)	0.001092 ± 0.00057	Chebyshev
heat	T-FNO (recursive)	0.001090 ± 0.00031	Fourier
wave1d	SNO (seq2seq)	0.001298 ± 0.00016	Fourier
wave1d_dir	sacheb_naive (seq2seq)	0.027387 ± 0.0011	Chebyshev
Burgers	T-FNO (recursive)	0.003963 ± 0.00098	Fourier
KdV	SNO (seq2seq)	0.000467 ± 0.000063	Fourier
wave2d	sacheb_naive (seq2seq)	0.007771 ± 0.0015	Chebyshev
wave3d	pureunif (recursive)	0.072375 ± 0.0089	Chebyshev
ns2d	sacheb_naive (seq2seq)	0.174500 ± 0.0014	Chebyshev

7.7 Supporting axes

Darcy (elliptic, Dirichlet). CKINO attains $8.7\times$ lower relative L^2 error than FNO overall and $18.3\times$ lower *at the boundary*, transferring zero-shot from 64^2 to 128^2 . This is the clearest non-periodic win for the Chebyshev basis.

Capacity ceilings. A 6k–400k sweep shows CKINO’s Burgers failure is *representational*, flat at ≈ 0.25 across a $60\times$ range, and its recursive KdV instability persists at every budget. Neither is fixed by scale.

Discretisation invariance. Training at $N = 64$ and evaluating at $N = 128$, SNO and T-FNO are essentially exactly invariant (0.99–1.00) *and* accurate. CKINO with a convolutional lift degrades by $57\times$ on advection; the spectral lift restores invariance (0.998) but costs one to two orders of magnitude of accuracy at the training grid ($8.2 \times 10^{-4} \rightarrow 0.475$). This is a genuine weakness of the Chebyshev construction, and SNO/T-FNO win the axis outright.

Inference speed. Operators beat the reference solver only where the solver is expensive: FNO is $3.2\times$ on Burgers, $7.7\times$ on KdV and $5.8\times$ on ns2d; CKINO is $2.3\times$ on KdV but $0.07\times$ on advection.

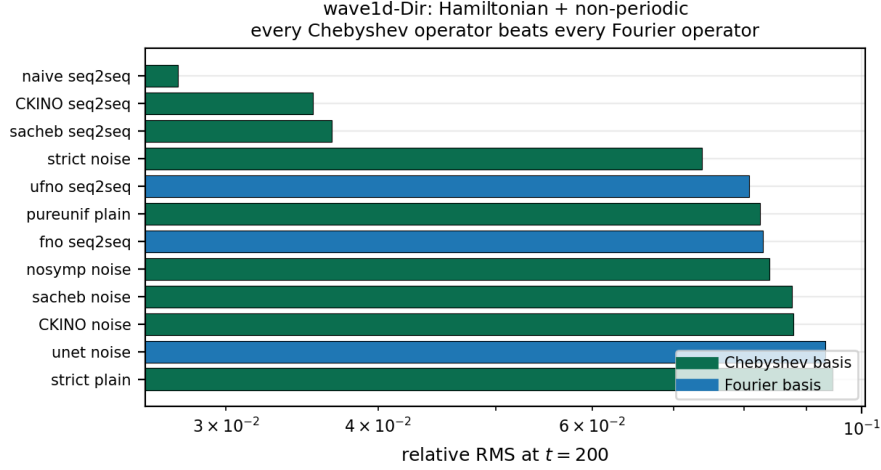


Figure 8: `wave1d_dir`, the Hamiltonian non-periodic rung. Every Chebyshev operator beats every Fourier operator: $3.0\times$ over the best FNO and $4.9\times$ over SNO, on a benchmark built to match SNO’s own data protocol. The mechanism is the Gibbs penalty the Dirichlet envelope imposes on a periodic basis.

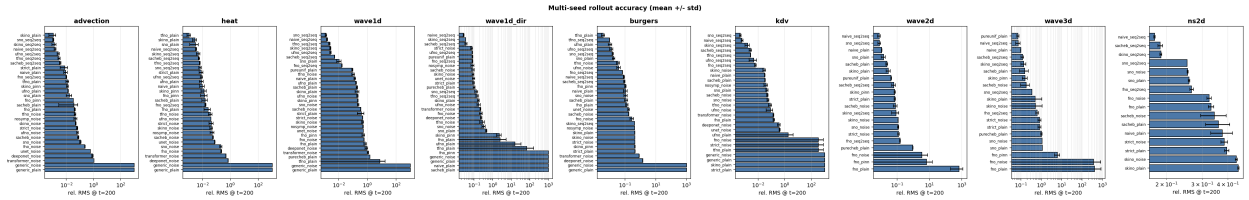


Figure 9: Relative RMS (log scale, mean $\pm$ std over 3 seeds) for every configuration, one panel per equation, all matched to $\approx 25k$ parameters.

8 Where each construction wins

9 Limitations

Re-implementations. SNO and GENERIC-FNO are our implementations from the published descriptions and were not tuned to the authors’ protocols.

Our GENERIC-FNO bug. It diverged in all 12 entries. Parameterising the PSD multiplier as $M = b^2$ with b initialised at zero places it at a saddle ($\partial M / \partial b = 2b = 0$), so the dissipative channel never activates. This is a defect of our implementation, *not* evidence about the published method, and no conclusion should be drawn from those rows.

Canonical structure only. The lift-free construction needs a canonical (q, p) pair and is therefore undefined on the scalar-field problems (advection, heat, Burgers, KdV, `ns2d`). KdV is Hamiltonian only under the non-canonical Gardner bracket, which we do not cover.

Scale and horizon. Domains are small (32 – 64 points in 1-D, 32^2 , 16^3 , 64^2), adequate to resolve the reported orderings but not a deployment benchmark. 2×10^4 steps separates bounded from unbounded but is still far short of the regimes where backward-error analysis is usually invoked.

Figure provenance. The field panels in Figures 11 and 12 are produced by a local reproduction with a reduced training schedule, because the GPU field archive did not survive download. They are qualitative; the orderings match the tables, the absolute errors are not the reported ones. All other figures come directly from the reported runs.

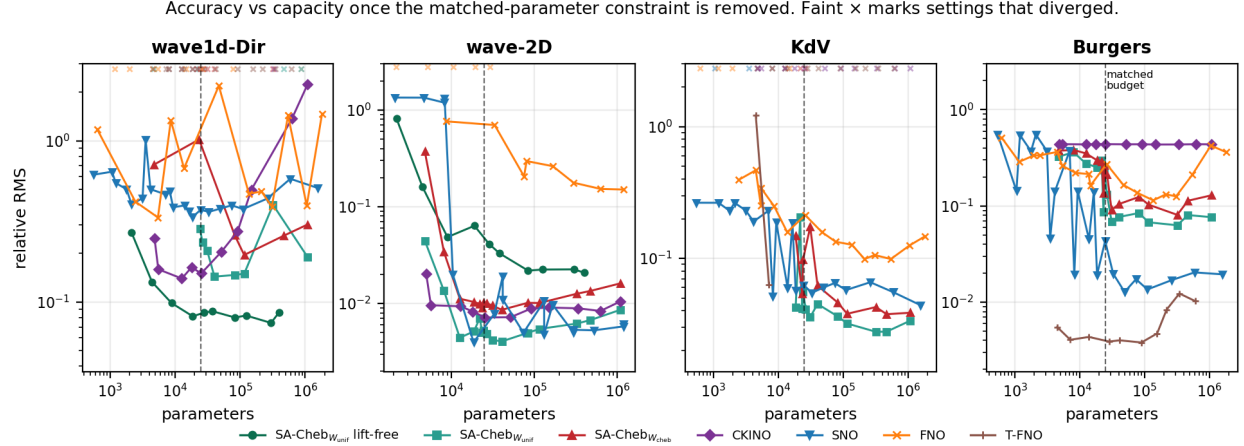


Figure 10: Accuracy against parameter count with the matched budget removed; the dashed line marks the 25k budget used in Table 4. The ranking is budget-dependent, so the matched table should be read as “best at 25k” rather than “best architecture”.

10 Conclusion

We set out to test whether geometric structure helps a learned operator and found that the question, as usually posed, is incomplete. Symplecticity is defined relative to an inner product; on a non-uniform grid two reasonable implementations preserve two different forms, both exactly, and only measurement distinguishes them. Once the form is named, the answer is clean: matching it to the grid wins 16 of 18 comparisons, and 4 of 4 by an order of magnitude more when the deployed map—not merely its core—carries the guarantee. That same construction is the only one in our study to survive a 2×10^4 -step rollout. Structure preservation pays; it is simply forfeited by the wrapper the field conventionally puts around it, and invisible unless you say with respect to what. We also retract a symplecticity theorem of our own that mistook unit Jacobian determinant for preservation of ω , and we think the general lesson is that such claims should be measured rather than argued—the measurement costs one line of autograd.

References

- M. Cranmer, S. Greydanus, S. Hoyer, P. Battaglia, D. Spergel, and S. Ho. Lagrangian Neural Networks. *ICLR 2020 Workshop on Integration of Deep Neural Models and Differential Equations*, 2020. arXiv:2003.04630.
- V. Duruisseaux, J. Kossaifi, and A. Anandkumar. Fourier Neural Operators Explained: A Practical Perspective. arXiv:2512.01421, 2026.
- M. Finzi, S. Stanton, P. Izmailov, and A. G. Wilson. Generalizing Convolutional Neural Networks for Equivariance to Lie Groups on Arbitrary Continuous Data. *ICML*, 2020. arXiv:2002.12880.
- S. Greydanus, M. Dzamba, and J. Yosinski. Hamiltonian Neural Networks. *NeurIPS*, 2019. arXiv:1906.01563.
- M. Grmela and H. C. Öttinger. Dynamics and thermodynamics of complex fluids. I. Development of a general formalism. *Physical Review E*, 56(6):6620–6632, 1997.
- E. Hairer, C. Lubich, and G. Wanner. *Geometric Numerical Integration: Structure-Preserving Algorithms for Ordinary Differential Equations*. Springer Series in Computational Mathematics, Vol. 31, 2nd edition, Springer, 2006.

Table 5: Honest summary. No single operator dominates; the axes separate.

construction	wins	loses
pureunif (lift-free, W_{unif})	long horizon (only bounded operator at 2×10^4 steps); wave3d at 25k; top of the unconstrained wave1d_dir sweep	needs a canonical (q, p) pair, so undefined on scalar fields; limited capacity per parameter
sacheb_naive (lifted, W_{unif})	wave1d_dir , wave2d , ns2d at 25k; best accuracy-per-parameter in 3-D	no long-horizon guarantee; diverges like the rest at 10^4 steps
CKINO	advection at 25k; advection/heat unconstrained by 10^2 – $10^3 \times$; Darcy by $8.7 \times$ ($18.3 \times$ at the boundary)	not symplectic in any form; Burgers ceiling; KdV instability; poor discretisation invariance
SNO (Makara et al., 2026)	wave1d , KdV; discretisation invariance; competitive in 2-D	Fourier basis mismatched on non-periodic domains ($4.9 \times$); expensive in 3-D
T-FNO (Kossaifi et al., 2023)	heat, Burgers; invariance	no structure; long-horizon divergence

- P. Jin, Z. Zhang, A. Zhu, Y. Tang, and G. E. Karniadakis. SympNets: Intrinsic structure-preserving symplectic networks for identifying Hamiltonian systems. *Neural Networks*, 132:166–179, 2020. arXiv:2001.03750.
- J. Kossaifi, N. Kovachki, K. Azizzadenesheli, and A. Anandkumar. Multi-Grid Tensorized Fourier Neural Operator for High-Resolution PDEs. arXiv:2310.00120, 2023.
- J. Kossaifi, N. Kovachki, Z. Li, D. Pitt, M. Liu-Schiaffini, R. J. George, B. Bonev, K. Azizzadenesheli, J. Berner, V. Duruisseaux, and A. Anandkumar. A Library for Learning Neural Operators. arXiv:2412.10354, 2024.
- N. Kovachki, Z. Li, B. Liu, K. Azizzadenesheli, K. Bhattacharya, A. Stuart, and A. Anandkumar. Neural Operator: Learning Maps Between Function Spaces with Applications to PDEs. *Journal of Machine Learning Research*, 24(89):4061–4157, 2023. arXiv:2108.08481.
- Z. Li, N. Kovachki, K. Azizzadenesheli, B. Liu, K. Bhattacharya, A. Stuart, and A. Anandkumar. Fourier Neural Operator for Parametric Partial Differential Equations. *ICLR*, 2021. arXiv:2010.08895.
- Z. Li, H. Zheng, N. Kovachki, D. Jin, H. Chen, B. Liu, K. Azizzadenesheli, and A. Anandkumar. Physics-Informed Neural Operator for Learning Partial Differential Equations. arXiv:2111.03794, 2023.
- L. Lu, P. Jin, and G. E. Karniadakis. Learning nonlinear operators via DeepONet based on the universal approximation theorem of operators. *Nature Machine Intelligence*, 3(3):218–229, 2021. arXiv:1910.03193.
- Y. Makara, Y. Tanaka, T. Matsubara, and T. Yaguchi. Symplectic Neural Operators for Learning Infinite Dimensional Hamiltonian Systems. arXiv:2605.15881, 2026.
- J. E. Marsden and T. S. Ratiu. *Introduction to Mechanics and Symmetry: A Basic Exposition of Classical Mechanical Systems*. Texts in Applied Mathematics, Vol. 17, 2nd edition, Springer, 1999.
- H. C. Öttinger. *Beyond Equilibrium Thermodynamics*. Wiley-Interscience, Hoboken, NJ, 2005.
- E. Perez, F. Strub, H. de Vries, V. Dumoulin, and A. Courville. FiLM: Visual Reasoning with a General Conditioning Layer. *AAAI*, 2018. arXiv:1709.07871.
- M. Raissi, P. Perdikaris, and G. E. Karniadakis. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics*, 378:686–707, 2019.

- J. Sulskis and S. Ravi. GENERIC-FNO: Embedding Energy Conservation and Entropy Production into Fourier Neural Operators. arXiv:2606.08343, 2026.
- L. N. Trefethen. *Spectral Methods in MATLAB*. SIAM, Philadelphia, 2000.
- A. J. Varghese, S. Liu, P. Chen, Y. Zhu, J. Darbon, and G. E. Karniadakis. PI-SONet: A Physics-Informed Symplectic Operator Network for Real-Time Optimal Control of Multi-Agent Systems. arXiv:2605.14332, 2026.
- A. Weinstein. Symplectic structures on Banach manifolds. *Bulletin of the American Mathematical Society*, 75:1040–1041, 1969.
- G. Wen, Z. Li, K. Azizzadenesheli, A. Anandkumar, and S. M. Benson. U-FNO—An enhanced Fourier neural operator-based deep-learning model for multiphase flow. *Advances in Water Resources*, 163:104180, 2022. arXiv:2109.03697.
- Z. Zhang, Y. Shin, and G. E. Karniadakis. GFINNs: GENERIC formalism informed neural networks for deterministic and stochastic dynamical systems. *Philosophical Transactions of the Royal Society A*, 380(2229):20210207, 2022.
- J. Zhang, Y. Wang, and G. Lin. LegONet: Plug-and-Play Structure-Preserving Neural Operator Blocks for Compositional PDE Learning. arXiv:2603.07882, 2026.
- Y. D. Zhong, B. Dey, and A. Chakraborty. Symplectic ODE-Net: Learning Hamiltonian Dynamics with Control. *ICLR*, 2020. arXiv:1909.12077.

A Qualitative field comparisons

B Retraction of a previously asserted guarantee

The CKINO operator was originally documented with an exact symplecticity theorem. We retract it, and record the error because it is instructive.

The withdrawn argument ran: each Störmer–Verlet substep has a Jacobian that is triangular with unit diagonal blocks, hence determinant 1, hence preserves ω . The step from “determinant 1” to “preserves ω ” is invalid. By Proposition 2, $\det D\Phi = 1$ holds for every shear, while preservation of ω_W additionally requires $WA = A^\top W$. The operator’s kernel (5) uses independent φ_r, ψ_r and an unconstrained channel mix, so A is not self-adjoint; the measured defect is ≈ 1.4 in *both* candidate forms (Figure 5). A corollary energy-drift bound, whose proof invoked the modified-Hamiltonian machinery of backward error analysis (Hairer et al., 2006), is retracted with it: that machinery requires symplecticity, and volume preservation alone does not imply the existence of a modified Hamiltonian.

A second, subtler retraction concerns the comparison itself. An earlier draft reported that exact symplecticity was a measurable cost, on the basis of a defect table computed against a single inner product. Scoring both variants against both forms (Figure 5) shows the “non-symplectic” control was exactly symplectic in the uniform form all along, and reverses the conclusion.

C The weighted adjoint in the low-rank basis

With $\alpha_r = \langle \psi_r, v \rangle_W$ and $(Kv)(x) = \sum_r \varphi_r(x) M_r \alpha_r$,

$$\langle Kv, g \rangle_W = \sum_r (M_r \alpha_r)^\top \langle \varphi_r, g \rangle_W = \sum_r \alpha_r^\top (M_r^\top \langle \varphi_r, g \rangle_W) = \left\langle v, \sum_r \psi_r M_r^\top \langle \varphi_r, g \rangle_W \right\rangle_W,$$

wave1d-Dir (Hamiltonian, Dirichlet): reference solver (grey) vs operator prediction, displacement field $u(x, t)$

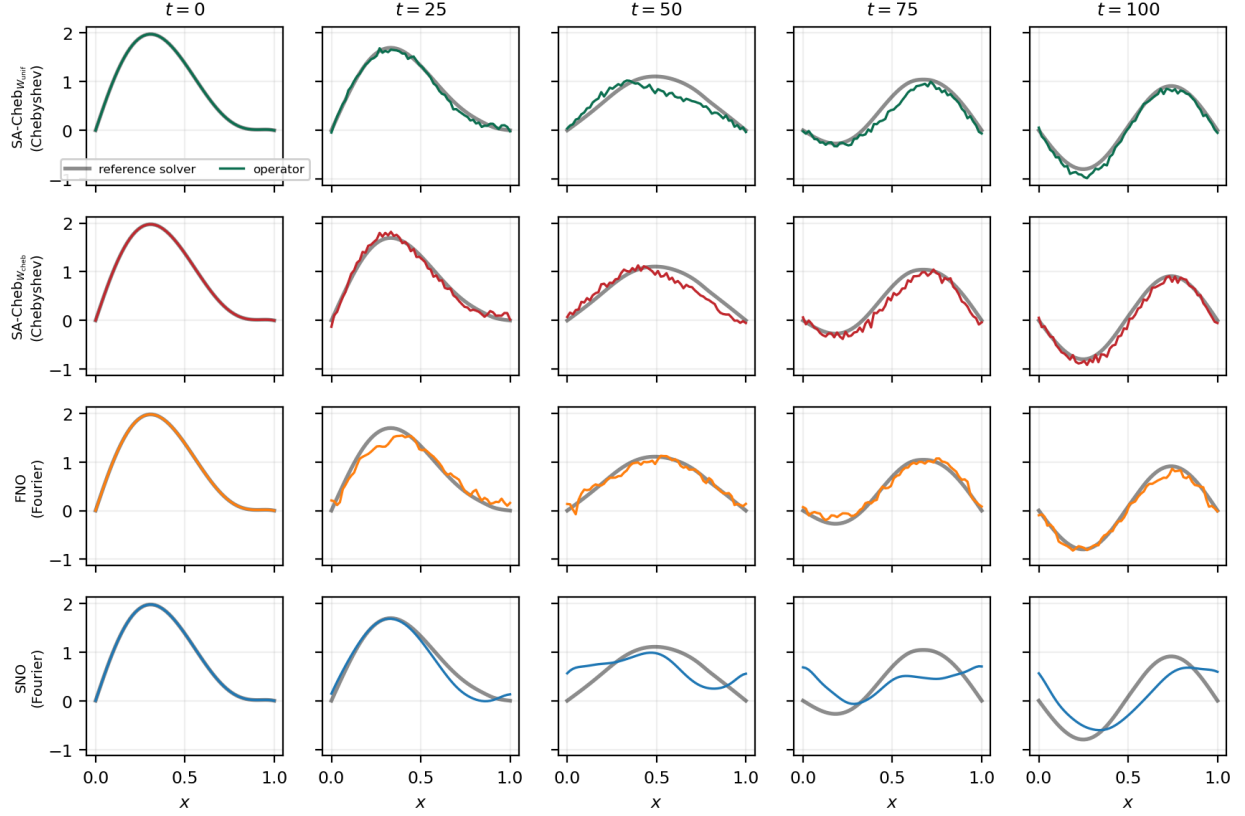


Figure 11: `wave1d_dir`: reference solver (grey) against operator prediction, displacement $u(x, t)$ at five rollout times. The Chebyshev operators track the Dirichlet boundary; the Fourier operators accumulate the Gibbs error that Figure 8 quantifies. Reduced-schedule local reproduction, see §9.

which is exactly $\langle v, K^*g \rangle_W$. Hence the weighted adjoint is obtained by exchanging φ and ψ and transposing the channel mixing; no inverse of W is formed and the cost remains $O(RN^d)$. In d dimensions the weight is the outer product of the per-axis weights and remains diagonal, so both this identity and Theorem 5 carry over unchanged.

wave1d, long autoregressive rollout: the lift-free operators stay on the solution manifold while FNO leaves it

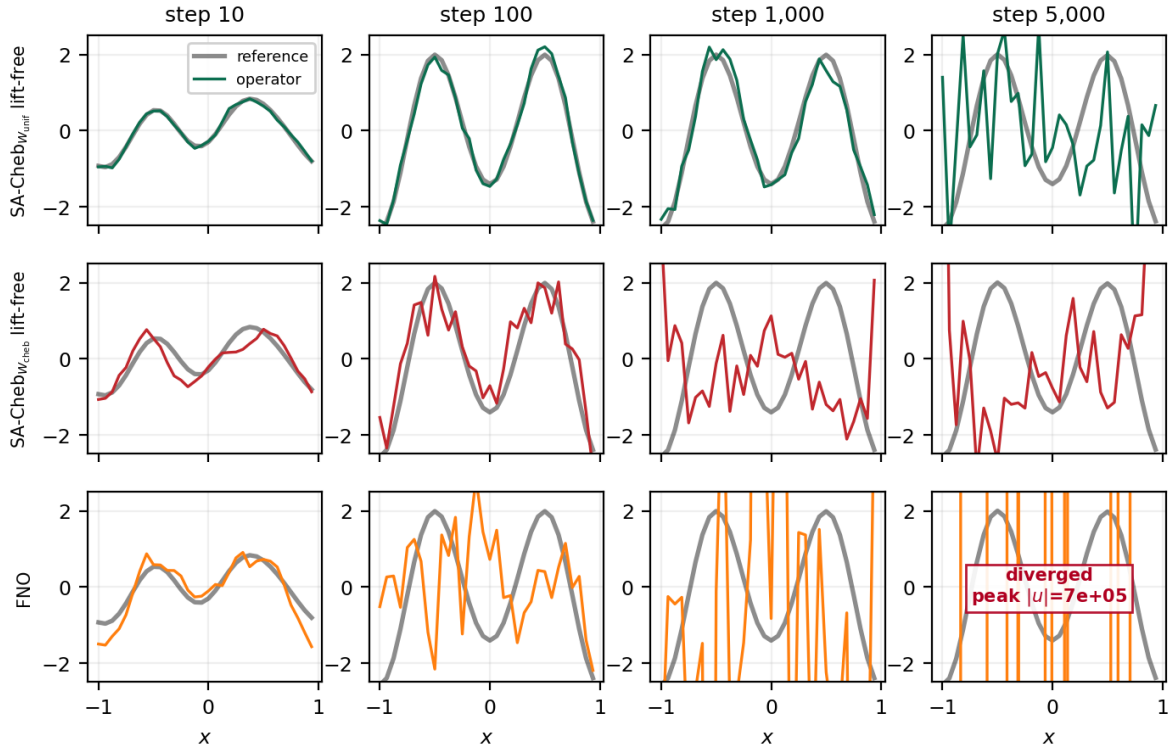


Figure 12: wave1d under long autoregressive rollout. The lift-free operators remain on the solution manifold at 5,000 steps while FNO has left it by 1,000. This is Figure 7 viewed as fields rather than curves. Reduced-schedule local reproduction, see §9.